



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0183



Agricultural Machinery – Biomass Furnace VSEE 5L-30 FURNACE

Test Report Number	2023 – 0183
Date of Issuance	December 1, 2023
Valid Until	December 1, 2028
Agricultural Machinery	Biomass Furnace
Type	Indirect-fired (Dry type), Tested using rice hull
Brand	VSEE
Model	5L-30 FURNACE
Test Requested by	GOLDENLEAF TRADING CORPORATION 1903 Meridian, Kasambagan, Cebu City

SUMMARY

Performance Criteria	Requirement as per PNS/PAES 242:2010	AMTEC Test
Biomass Fuel Consumption, kg/h	40-100 ^a	113.83 ^d 137.56 ^e
Allowable Operating Temperature, °C, maximum	ND ^a	62.0 ^d 64.5 ^e
Burning Efficiency, %, minimum	95	99.32
Furnace Efficiency, %, minimum	65	NM ^f
Heating System Efficiency, %, minimum	50 ^b	53.63 ^d 16.91 ^e
Flue Gas Emitted, mg/Ncm, maximum	35 ^c	NM ^g

^a Data were based on submitted brochure. The requesting party did not submit complete machine specifications.

^b Requirement for indirect-fired biomass furnace based on PNS/PAES 242:2010.

^c Allowable level of pollutants as required by the Clean Air Act of the Philippines.

^d Data collected during the dryer operation of VSEE 5H-12 Dryer.

^e Data collected during the dryer operation of VSEE 5B-6 Dryer.

^f The parameter cannot be measured due to the inaccessibility of the component.

^g There is no available instrument to facilitate the measurement of the parameter during the test operation.

Note: ND – No Data, NM – Not Measured

OBSERVATIONS

1. Operator's Manual	A general operator's manual biomass furnace was provided but not specific to the model tested.
2. Previous test reports of similar brand and model	None
3. Manufacturer	ANHUI VSEE OPTOELECTRONIC TECHNOLOGY CO. 659 Ming Zhu Road, High Tech Zone, He Fei City, Anhui Province, China
4. Prime mover used	0.15 kW Electric Motor (Screw feeder), 0.37 kW Electric Motor (Main blower), 2.2 kW Electric motor (Combustion blower), 0.37 kW Electric motor (Ash auger)
5. Number of operators	Two (One feeding operator/discharge collector and one machine operator)
6. Cracking/scaling on any part of the furnace	There is no cracking/scaling observed on the furnace.
7. Adjusting and repairing of parts	The temperature and blower opening can be adjusted. The furnace has an automatic control system that regulates the set temperature by automating the feeding rate of the hull and blower function. Locally fabricated parts can be repaired. No repairs were observed during the test.
8. Cleaning of parts	Some parts such as the furnace chamber, hull feeding unit, and ash discharge unit are accessible for cleaning.
9. Transporting the furnace	The furnace is stationary.
10. Availability of switches	Start/stop buttons, alarm, and touchscreen display were provided on the control box.
11. Safety features	All rotating parts were provided with covers. The combustion chamber was properly enclosed during operation.
12. Shall have combustion chamber and wall with heat-resisting lining, insulation, steel supporting structure and casing, hopper with feeding mechanism, ash discharge unit, ash arrester, heat exchanger (for indirect-fired), and flue gas chimney (for indirect-fired).	The furnace is composed of a combustion chamber with walls of heat-resisting lining, a steel supporting structure, a hopper with a screw-type conveyor feeding mechanism, and a screw-type ash discharge unit and flue gas cyclone.

OBSERVATIONS (Cont'n)

13. Single-layer heat-resisting lining of the combustion chamber and walls shall be able to withstand the operating temperature of at most 760°C.	The maximum operating temperature inside the combusted chamber and wall was not verified due to the lack of a thermocouple applicable for higher temperatures.
14. Biomass furnace that is operating at higher temperature than 760°C shall have a multi-layer wall of hearth.	The type of wall hearth cannot be verified due to installation.
15. Multi-layered lining shall be composed of two or three layers of refractory and an insulator.	The composition of the lining cannot be verified due to installation..
16. The heat-resisting lining of the combustion chamber and walls of the biomass furnace shall be made of individual bricks, monolithic linings, and/or ceramic fiber lining.	The type of lining cannot be verified due to installation.
17. Brick lining shall be made of fire brick from fireclay and kaolin or silica brick. a. The brick shall be staggered to avoid a direct gas path from the hot gas side to the shell. b. Bricks shall be assembled using high-temperature mortar based on high alumina cement. c. Ceramic fiber board or insulating brick shall be used for multi-layered brick lining. d. Expansion joints between the bricks shall be provided at regular intervals to prevent overstressing the brick during operation.	The composition of the lining, as well as the presence of high-temperature mortar, ceramic fiber board or insulating brick, and expansion joints, were not verified since the chamber was already installed prior to the test.

OBSERVATIONS (Cont'n)

18. Monolithic lining shall be made of refractory materials graded to a range of sizes together with bonding agent attached to anchors. a. For castable refractory, anchors shall be welded or bolted to the furnace shell. The anchors shall be made of low thermal expansion steel metal (e.g. chromium-base steel, tungsten-base steel, molybdenum-base steel) and/or high-temperature stainless steel. b. Castable refractory shall be made of heat-resisting aggregates and alumina cement that can be poured into forms. c. Shall be made into a gas-tight structure to avoid heat loss.	The composition of the lining cannot be verified due to installation.
19. Ceramic fiber lining shall be made of ceramic fiber with studs or proprietary fixings welded to steel shell.	The composition of the lining cannot be verified due to installation.
20. Insulation shall be made of fireproof materials like ceramic fiber or ceramic wool, 85% magnesia, rock wool, and/or mineral wool. It shall be placed between the wall lining and the steel frame of the biomass surface.	The type of insulation cannot be verified due to the inaccessibility of the installed parts.
21. Supporting structure and casing shall compose of steel shell or plate work and load-bearing frame.	The supporting structure and casing are made of steel plates and angle bars. The frame was able to withstand the load of the furnace.
22. The supporting structure and casing shall be able to support the entire load of the biomass furnace during operation.	The supporting structure and casing were able to support the entire load of the biomass furnace during operation.
23. Shall be painted with light color paint or aluminum paint	The supporting structure and casing of the furnace chamber were painted blue. The heat exchanger supporting structure and casing were not painted but were made of stainless steel sheets.
24. Hopper should be made of metal sheet. The angle of hopper shall conform to the angle of repose of the biomass fuel (30°-36° for corncob; 35°-50° for rice hull)	The hopper is made of metal sheets. The angle of repose was not verified due to inaccessibility.

OBSERVATIONS (Cont'n)

25. Hopper should have a feeding section (e.g., screw feeder, rotary, etc.) to convey the biomass fuel directly to the combustion chamber.	The hopper of the furnace is equipped with a screw feeder for rice hull feeding.
26. Combustion chamber shall have a force draft blower and/or passages/openings for air inlet to ensure the complete burning of biomass fuel. For rice hull furnace, this blower or chimney shall be able to support combustion and sustain temperature of 700°C.	A blower is used to introduce air into the combustion chamber. The capability of the design to sustain a combustion temperature of 700°C was not verified due to limited technical capability onsite.
27. Wet-type indirect-fired biomass furnace shall have induced draft either by blower or chimney.	Not applicable since the furnace is dry-type indirect-fired.
28. For indirect-fired biomass furnace, fire tube for dry-type furnace and boiler tube for wet-type furnace shall be used in heat exchanger.	Fire tubes were used in the heat exchanger.
29. The center-to-center distance of the heat exchanger tubes installed shall be more than 1.25 times the tubes' outside diameter.	The center-to-center distance of the heat exchanger tubes was not verified due to inaccessibility.
30. Total cross-sectional area of the heat exchanger used in biomass furnace shall be equal to the cross-sectional area of the duct connected to the plenum used in grain dryer.	The total cross-sectional area of the heat exchanger was not verified due to installation.
31. Tube layout for heat exchanger should be triangular patterns.	Tubes in the heat exchanger were not verified due to installation.
32. Tubes of heat exchanger shall have provision for ease of replacement.	The tubes for the heat exchanger were incorporated above the combustion chamber.
33. Grate for grate-type biomass furnace shall be made of flat or square high temperature resistant stainless steel and/or high temperature steel bars such as chromium-base steel, tungsten-base steel, and molybdenum-base welded together.	Not applicable
34. There should be a provision for collecting and discharging ash.	A screw conveyor was used as an ash-discharging mechanism and ashes were collected using a wheelbarrow.

OBSERVATIONS (Cont'n)

35. Biomass furnace (indirect-fired) shall have a burning fly ash collector such as ash arrester, ash bin, and/or scrubber system.	It was not verified if burning fly ash collector was provided due to installation.
36. Bolts and screws to be used shall conform to the requirements of PAES 311 and 313.	The bolts and screws used were not verified if they were conforming to the requirements of PAES 311 and 313 due to limited technical capability onsite.
37. Failure or abnormalities that may be observed on the furnace or its component parts during and after the operation	None
38. Other remarks	a. The biomass furnace was tested while providing heat simultaneously for the drying operation of VSEE 5H-12 Heated-Air Mechanical Grain Dryer & VSEE 5B-6 Heated-Air Mechanical Grain Dryer. b. The nameplate and warning notices on the biomass furnace were not provided at the time of the test.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Dates of Test	April 4-5, 2023
Location of Test	San Isidro, St. Bernard, Southern Leyte 10°17'30.11" N, 125°07'53.46" E
Items Tested	Operator's Manual, Burning Efficiency, Heating System Efficiency, Rice Hull Consumed, Rice Hull Consumption Rate, Power Consumption, and Number of Operators

METHODS OF TEST

The biomass furnace was tested based on PNS/PAES 243:2010 – Agricultural Machinery – Biomass Furnace – Methods of Test.

DESCRIPTION OF THE MACHINE

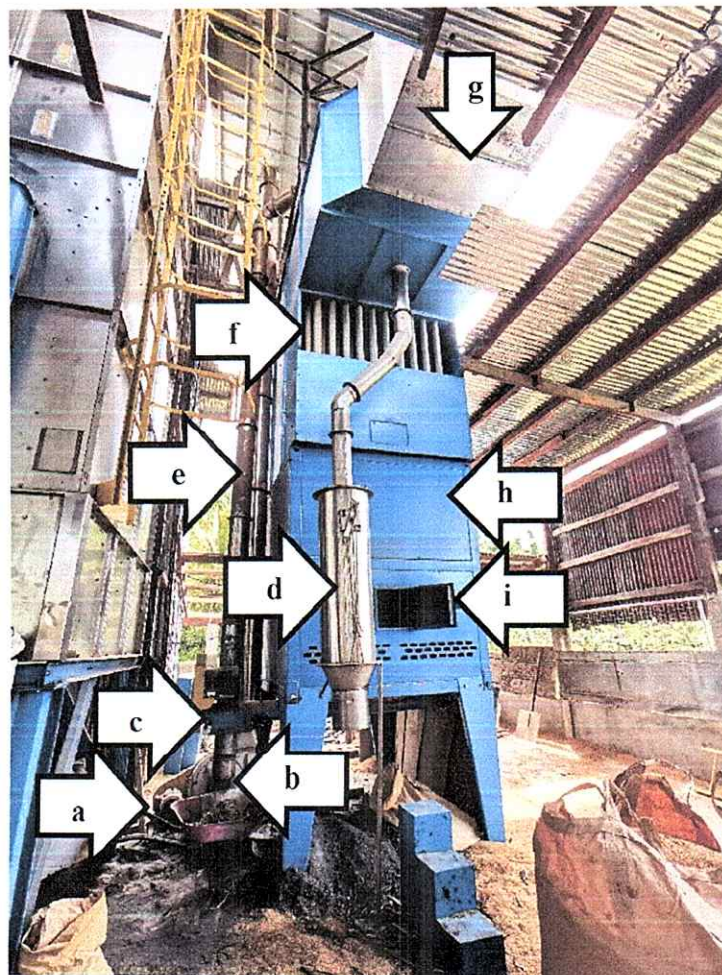


Figure 1. Parts of the VSEE 5L-30 FURNACE Biomass Furnace include the following:

- (a) Prime mover of the main blower, (b) Ash outlet, (c) Ash discharge unit,
- (d) Flue gas cyclone, (e) Chimney, (f) Heat exchanger chamber,
- (g) Air duct to dryer, (h) Combustion chamber, and
- (i) Combustion chamber's access door.

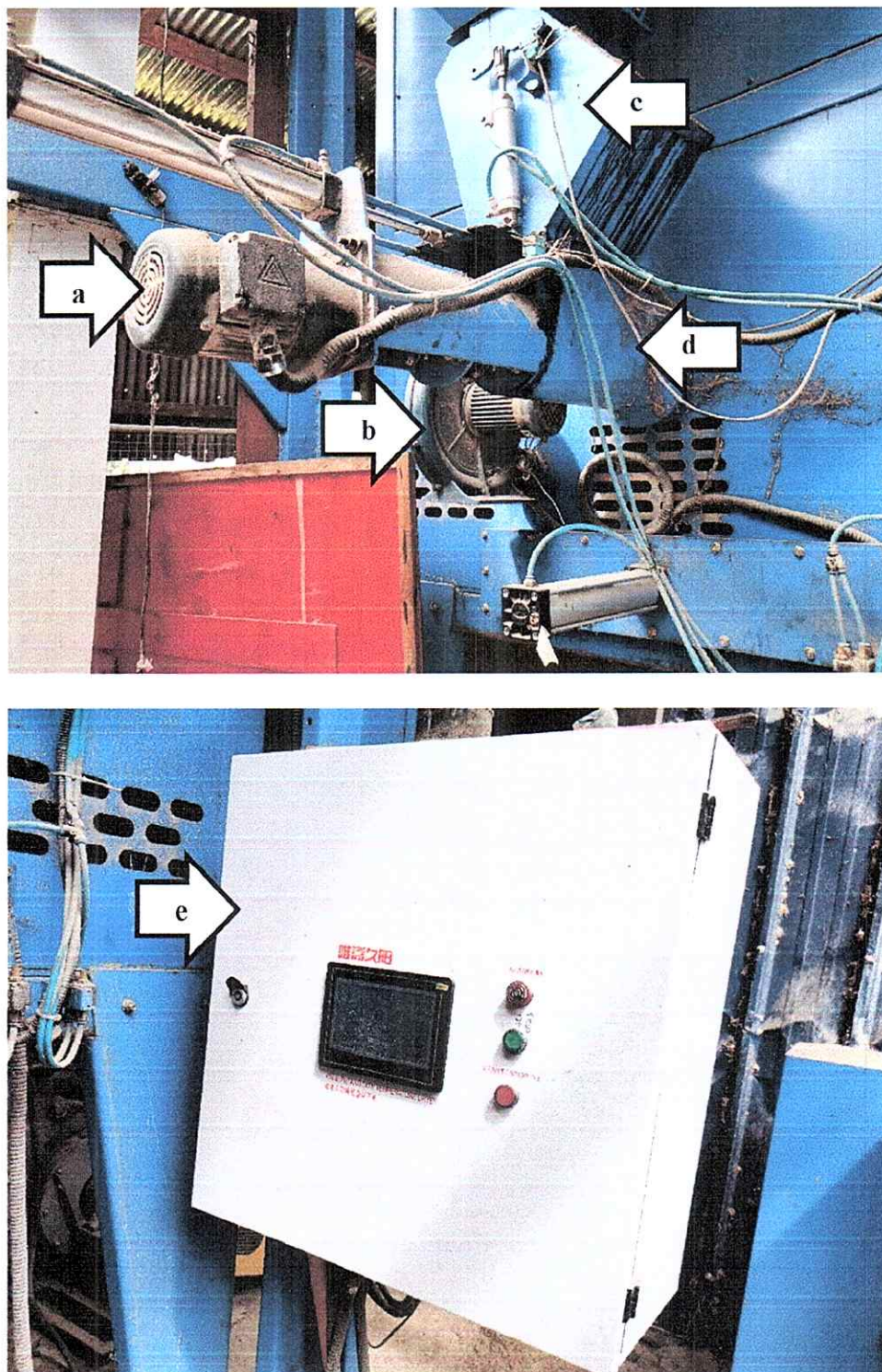
DESCRIPTION OF THE MACHINE (Cont'n)

Figure 2. Parts of the VSEE 5L-30 FURNACE Biomass Furnace include the following:
(a) Prime mover of screw feeder, (b) Combustion blower, (c) Input hopper,
(d) Feeding auger mechanism, and (e) Control box.

SPECIFICATIONS

Item		Manufacturer's Specification ^a	AMTEC Verification
1	Hopper		
1.1	Materials of construction	ND	Metal sheet
1.2	Number	ND	1
1.3	Dimensions, L × W × H, mm	ND	NM ^b
1.4	Screw feeder		
1.4.1	Dimensions, D × L, mm	ND	172 × 770
1.4.1.1	Thread thickness, mm	ND	NM ^b
1.4.1.2	Thread height, mm	ND	NM ^b
1.4.2	Prime mover		
1.4.2.1	Screw feeder prime mover		
1.4.2.1.1	Brand	ND	ND
1.4.2.1.2	Model	ND	ND
1.4.2.1.3	Serial number	ND	ND
1.4.2.1.4	Type	ND	ND
1.4.2.1.5	Rated power, kW	ND	ND
1.4.2.1.6	Rated speed, rpm	ND	ND
1.4.2.1.7	Voltage, V	ND	ND
1.4.2.1.8	Current, A	ND	ND
1.4.2.1.9	Frequency, Hz	ND	ND
1.4.2.1.10	Electric service required	ND	ND
1.4.2.1.11	Weight, kg	ND	ND
2	Combustion chamber		
2.1	Frame		
2.1.1	Materials of construction	ND	Steel bar and sheet
2.1.2	Dimensions, L × W × t, mm	ND	2710 × 1830 × NM ^b
2.2	Walls		
2.2.1	Materials of construction	ND	Steel bar and sheet
2.2.2	Dimensions, L × W × t, mm	ND	2710 × 1830 × NM ^b
2.2.3	Number of linings	ND	ND ^b
2.2.4	Insulation material used	ND	ND ^b
2.3	Combustion air inlet device used	ND	Blower

^a The requesting party did not submit manufacturer's specifications sheet.

^b The part is inaccessible due to installation.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
3	Heat exchanger		
3.1	Type of furnace	ND	Indirect
3.2	Tubes		
3.2.1	Materials of construction	ND	NM ^b
3.2.2	Dimensions, L × D × t, mm	ND	NM ^b
3.2.3	Orientation	ND	Vertical
4	Chimney		
4.1	Materials of construction	ND	Stainless steel sheet
4.2	Dimensions, L × D × t, mm	ND	NM ^b
5	Plenum		
5.1	Materials of construction	ND	Stainless steel sheet
5.2	Dimensions, L × D × t, mm	ND	NM ^b
6	Ash discharge unit		
6.1	Type	ND	Screw conveyor
6.2	Materials of construction	ND	Steel
6.3	Dimensions, D × L, mm	ND	172 × 2230
6.4	Prime mover		
6.4.1	Brand	ND	ND
6.4.2	Model	ND	ND
6.4.3	Serial number	ND	ND
6.4.4	Type	ND	ND
6.4.5	Rated power, kW	ND	ND
6.4.6	Rated speed, rpm	ND	ND
6.4.7	Voltage, V	ND	ND
6.4.8	Current, A	ND	ND
6.4.9	Frequency, Hz	ND	ND
6.4.10	Electric service required	ND	ND
6.4.11	Weight, kg	ND	ND
6.5	Ash pan	ND	ND ^b
6.6	Ash arrester		
6.6.1	Materials of construction	ND	NM ^b
6.6.2	Dimensions, H × D	ND	NM ^b
6.6.3	Maximum capacity, kg	ND	NM ^b

^a The requesting party did not submit manufacturer's specifications sheet.

^b The part is inaccessible during verification.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification^a	AMTEC Verification
7	Fans		
7.1	Materials of construction	ND	Steel sheet
7.2	Diameter of fan wheel	ND	NM ^b
7.3	Flow rate, m ³ /min	ND	NM ^b
7.4	Static pressure, mmH ₂ O	ND	NM ^b
7.5	Fan air speed, m/s	ND	NM ^b
8	Safety features	ND	Start and stop button with alarm
9	Other special features	ND	Automatic screw feeder and temperature control with touchscreen display; Can supply heat to 2 dryers simultaneously.

^a The requesting party did not submit the complete manufacturer's specifications.

^b The part is inaccessible during verification.

Note: ND – No Data, NM – Not Measured

RESULTS

Item		Data	
1	Information on biomass fuel		
1.1	Type	Rice hull	
1.2	Weight consumed, kg	1238 ^a 1366 ^b	
1.3	Moisture content, % _{wb}	7.03	
1.4	Bulk density, kg/m ³	110.37	
1.5	Heating value of biomass material, kJ/kg	13927	
2	Performance test	VSEE 5B-6 Heated Air Mechanical Grain Dryer	VSEE 5H-12 Heated Air Mechanical Grain Dryer
2.1	Ambient air temperature, °C		
2.1.1	Wet bulb	25.6	25.3
2.1.2	Dry bulb	27.7	27.3
2.2	Ambient air relative humidity, %	84.2	84.9
2.3	Air flow rate, m ³ /min	210.95	449.73
2.4	Heated air temperature from furnace, °C	NM ^c	NM ^c
2.5	Drying air temperature, °C	51.0	56.5
2.6	Total operating time, h	9.00	12.00
2.7	Furnace firing time, h	0.250	
2.8	Biomass fuel consumption rate, kg/h	137.56	113.83
2.9	Weight of ash residue sample after operation, g	4.29	
2.10	Weight of ash residue sample after further burning in electric furnace, g	4.17	
2.11	Weight of biomass material before complete burning in electric furnace, g	3.05	
2.12	Weight of ash after completely burning the biomass material in electric furnace, g	0.588	
2.13	Percentage of ash residue in the sample collected from the biomass furnace, % _{db}	0.198	
2.14	Percentage of ash in biomass material after complete burning in electric furnace	0.193	

^a Total rice hull consumed during simultaneous drying operations until VSEE 5B-6 Heated-Air Mechanical Grain Dryer completed the drying operation.

^b Total rice hull consumed until completion of the whole drying operation of the two heated-air mechanical grain dryers.

^c Not measured due to inaccessibility during operation.

Note: NM – Not Measured

RESULTS (Cont'n)

Item		VSEE 5B-6 Heated Air Mechanical Grain Dryer	VSEE 5H-12 Heated Air Mechanical Grain Dryer
2.15	Heat available in furnace, kJ/h	17241530.73	19024186.73
2.16	Volume flow rate of air, m ³ /h	12656.96	26983.55
2.17	Mass of ash collected after burning in biomass furnace, kg	116.5	152.0
2.18	Fan air velocity, m/s	16.41	11.75
3	Biomass furnace performance		
3.1	Hopper load capacity, kg	NM ^a	
3.2	Heat transferred at the heat exchanger, kJ/h	446848.97	896536.95
3.3	Heated air temperature at the heat exchanger, °C	64.5	62.0
3.4	Heat released by ash residue sample per kg of ash residue, kJ/kg	0.818	0.627
3.5	Heating system efficiency, %	16.91	53.63
3.6	Burning efficiency, %	99.32	
3.7	Furnace efficiency, %	NM ^b	NM ^b

^a Not measured due to inaccessibility during operation.

^b Not measured due to inaccessibility of parts for temperature measurement.

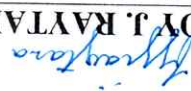
Note: NM – Not Measured

OBSERVATION



Figure 3. Markings and labels on the control panel of the VSEE 5L-30 FURNACE Biomass Furnace.

Tested by:


FATIMA JOY J. RAYTANA
Test Engineer

Verified by:


MARIE JEHOA B. REYES
Test Engineer

Approved for Release:


ARTHUR L. FAJARDO
AMTEC Director

Date

AUG 24 2023

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